Scope, Related Work, and Positioning

Scope, Related Work, and Positioning I

This section situates the exemplar scientifically and states explicit boundaries. The goal is not to compete with monographs on nonlinear programming [@nocedal2006numerical;@bertsekas1999nonlinear], but to show how a minimal, test-backed optimization story fits the template's reproducibility and rendering stack [@peng2011reproducible].

Classical gradient methods I

Smooth unconstrained minimization via first-order updates has a long lineage, from Cauchy's early descent perspective [@cauchy1847methode] to modern treatments of convex problems [@boyd2004convex] and accelerated gradient schemes [@nesterov2013gradient]. Polyak's classical discussion of gradient convergence factors remains relevant when interpreting empirical iteration counts [@polyak1964some]. The present manuscript restricts attention to **fixed-step** gradient descent on a **convex quadratic**, where rates reduce to scalar linear recurrences in the error ([@sec:results], [@eq:scalar_linear_update]).

Adaptive and stochastic extensions I

Practical machine-learning optimizers (e.g., Adam [@kingma2014adam]) introduce momentum, adaptive preconditioning, or noise from minibatching. Those methods are **out of scope** for `template_code_project`: the exemplar deliberately keeps the algorithm minimal so that failures (divergent α, iteration caps) are interpretable without confounding from stochastic sampling or line-search logic.

What this project proves about the template I

The scientific claims through [@sec:introduction], [@sec:methodology], and [@sec:results] are standard textbook material. The **non-standard** contribution is procedural: configuration in `manuscript/config.yaml` drives `run_convergence_experiment()`, figures, CSV exports, and `{{RESULT_*}}` substitution (`scripts/z_generate_manuscript_variables.py`) so that PDF, HTML, and validation logs refer to the same numbers. That pattern is what downstream projects should copy—whether the domain is optimization, differential equations, or Bayesian workflows.

Explicit limitations I

1. **Dimensionality**: Default experiments emphasize $d = 1$ with $A = I$ for transparent plotting; the benchmark figure explores $d > 1$ only with identity Hessians, so no ill-conditioning effects appear.
2. **Step-size policy**: Only constant α is implemented in `src/optimizer.py`; there is no Wolfe or Armijo backtracking.
3. **Global optimization**: Convexity is assumed; no basin-hopping or restarts are studied.
4. **Numerical model**: Double-precision floating point only; no interval or arbitrary-precision analysis.

These limitations are intentional: they narrow the failure surface so that infrastructure concerns—tests, logging, figure registration, and PDF cross-references—remain visible rather than buried under algorithmic complexity.